



The
University of
Faisalabad

Dynamic Packet Filters

Lecture 18



- **Dynamic Packet Filters** are smart firewalls that **keep track of active connections** and **adjust their rules automatically** based on the state of those connections.
- They're also known as **stateful firewalls**.

How They Work (in simple terms):

- When a computer on your network starts a connection (like opening a website), the firewall **remembers it**.
- It creates a “rule” that allows traffic **only for that connection**.
- When the connection ends, the rule **disappears automatically**.
- This makes the firewall **dynamic** — it changes with the traffic.

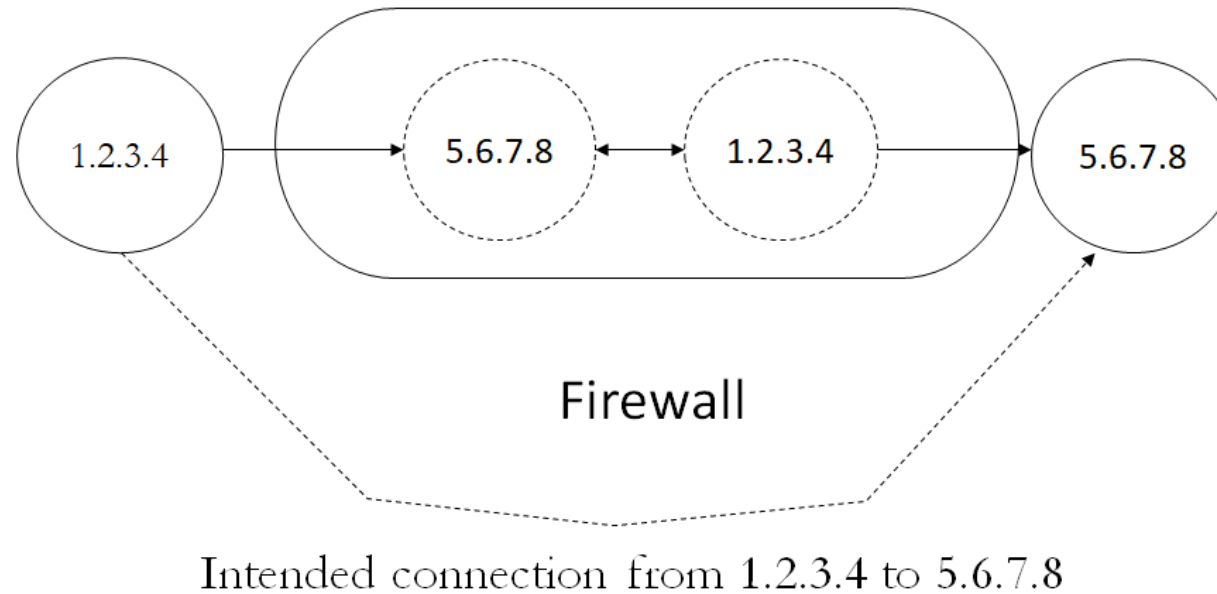
Why It's Useful

- **Protects your network:** Only traffic that's part of an allowed connection is permitted.
- **Transparent to users:** You don't need to manually allow every packet.
- **Smart rules:** It understands what belongs together (e.g., a request and its response).

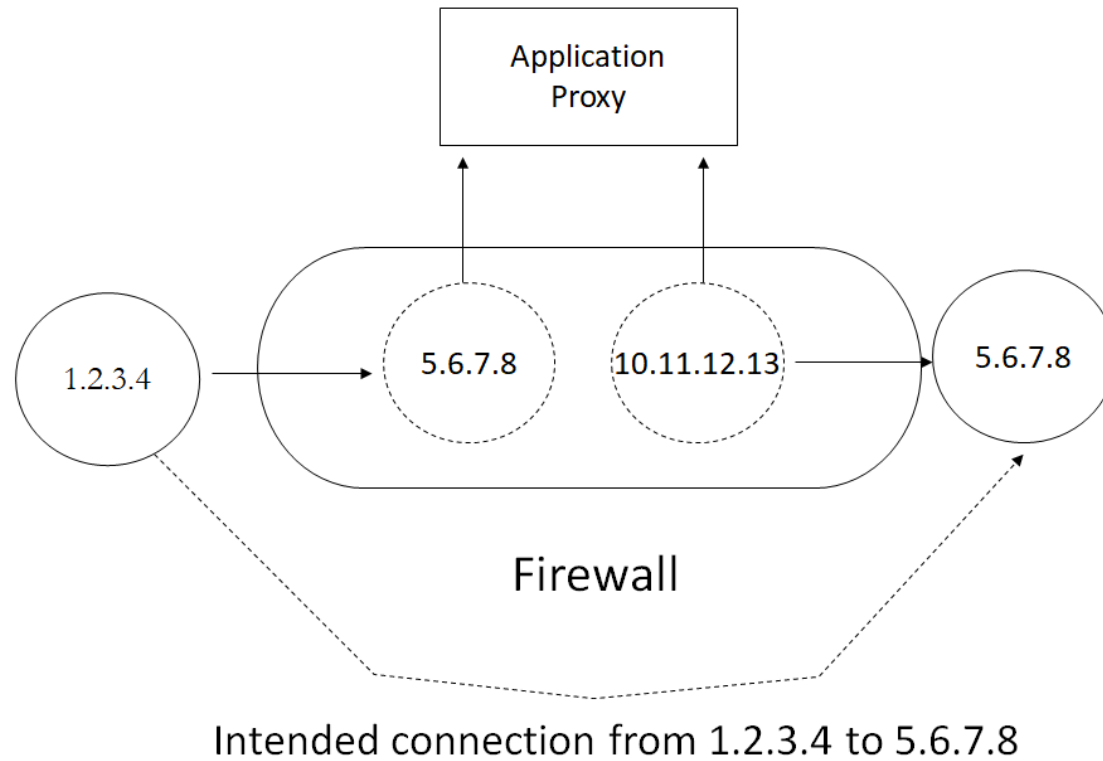
- **"Captures semantics of a connection" — What does that mean?**
- It means the firewall **understands how real connections work** — like knowing:
 - A connection starts with a request (e.g., "Can I connect?")
 - Then comes a response (e.g., "Yes, here's the data")
 - Then it ends (e.g., "Goodbye")
- So it can **match packets to these steps**, and only allow traffic that makes sense.

Benefits

- **Good protection** for administrators
- **Full transparency:** Users don't see delays or blocking unless something suspicious happens
- **Automatic rule updates:** No need to constantly write manual rules
- **Better control:** The network has full control over which sessions are allowed



Redialing on a dynamic packet filter. The dashed arrow shows the intended connection; the solid arrows show the actual connections, to and from the relay in the firewall box. The Firewall impersonates each endpoint to the other.



A dynamic packet filter with an application proxy. Note the change in source address

1. Attacks from the Inside

- Firewalls are designed to block **outside threats**, not **people already inside**.
- If someone inside your network is an “evildoer” (a bad actor), they can do harm **without needing to go through the firewall**.

2. Malicious Code on Internal Machines

- If a user **accidentally runs a virus or malware** from an email or USB, the firewall won't know.
- The **attack starts inside** the network, so it can **bypass the firewall**.

3. High-Risk Organizations

- Some places like **banks, military, or government agencies** have more insider threats.
- That means they need **extra layers of protection**, not just firewalls.

4. Protection Must Exist at Every Layer

- Firewalls are only one part of the defense. You also need:
- Antivirus software
- Employee training
- Secure access controls
- Intrusion detection systems (IDS)
- Encryption

5. Can't Catch Every Virus or Bad File

- There are **millions of file types and operating systems (Windows, Mac, Linux, etc.)**.
- It's **impossible** for a firewall to recognize **every possible infected file**.

What Should Be Done Instead?

- Think in **layers**: Don't rely on just one tool.
- **Assess risks at every level** — from the network, to the devices, to the people using them.
- Use the rule: “**Defense in depth**” — multiple security measures working together.

- **How Many Routers Do We Need?**
- If routers **only support outgoing filtering**, then you need **two routers**:
- **First Router:**
 - Filters **outgoing traffic**
 - Protects against **compromised gateway machines**
- **Second Router:**
 - Filters **outgoing traffic from the internal network**
 - Stops **address forgery** and controls access to the **gateway**

Routing Filters

- Routing filters help **control where data can go** in a network.

Why It Matters

- Normally, every device on the Internet is **reachable**.
- But we want to **protect sensitive parts** of our network.
- **Routers** can **hide or block certain paths** using routing filters.

What's the Risk?

- **IP Source Routing** lets a sender pick the route their data takes.
- Hackers can use this to **bypass security** and reach "unreachable" or protected machines.
- They can also use it for **address spoofing** (pretending to be someone else).

What's the Fix?

- **Block source routing** at the **edges** (border routers), not inside the core (backbone).
- This stops attacks before they get in.

- Packet filters make route filters less necessary by blocking unwanted traffic directly. Route filtering gets tricky in complex networks. Sometimes, networks use IPs that belong to others (route squatting), which can cause issues. It's also hard to find unused IPs. Packet filters are a simpler, safer solution.

Question?